Essay 2: Cyber Coercion Against Ukraine: A Russian Military Advisor's Assessment

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1 Introduction

Russian cyber operations against Ukraine have continued for more than 10 years, largely coinciding with political events and kinetic conflicts such as the 2014 and 2022 invasions. Cyber operations such as the 2015 and 2016 power grid attacks or the 2017 NotPetya worm imposed significant material and financial costs, but yielded few if any political concessions from Ukraine. However, at the heart of Russia's military doctrine, "New Generation Warfare" or the "Gerasimov Doctrine" is the usage of cyber operations in a supporting manner for kinetic operations rather than as an independent, standalone coercive tool. However, this raises an analytically useful question: if Russia were to design an optimal standalone cyber campaign to compel Ukrainian surrender, what would it look like, and could it succeed?

Answering this question requires distinguishing between coercion and brute force. As Biddle (2020) explains, coercion (specifically compellence) aims to change an adversary's will, making resistance appear to be more costly than concession, rather than physically denying the adversary the capacity to fight. A coercing power must make resistance appear to be more costly than concessions. In today's current digital society, this is more prominent than ever given the reliance of modern economies, government, and societies on cyber and broader Information Control Technology (ICT). From this point of view, any operation

that disrupts critical cyber or ICT technology and infrastructure has the capacity to inflict great harm and damage. However, what is unknown is whether those costs can ever be severe, credible, and sustained enough to break an adversary's resolve. A typical barrier to cyber coercion identified in relevant literature is the issue of attribution, as a target cannot be certain of the identity of the attacker, making its cost-benefit calculation more difficult (Gartzke 2013). Given the current hostilities, attribution is generally a non factor. This essay therefore sets aside attribution and focuses on the structural and informational barriers to cyber coercion.

However, this framework is partially incomplete. Coercion theory primarily focuses on a rational cost benefit calculation, which risks understating the importance of ideology and morale in shaping an adversary's willingness to resist. A state whose population views the conflict as existential or ideologically necessary rather than a policy may prove immune to cost imposition regardless of the severity of said cost.

Many cyber arguments suggest that it is not possible, but similar parallels of coercion with kinetic means demonstrate non rational cost-benefit analysis. Take, for instance, Iran's position in operation Epic Fury. The United States and Israel devastated Iran's military infrastructure, destroyed its ballistic missile arsenal and navy, and reportedly killed its supreme leader. However, the regime continued to retaliate across the Gulf and refused to capitulate, sustained by revolutionary ideology and a framing of the conflict as existential (CSIS 2026). If overwhelming kinetic force across the full spectrum of a state's military and leadership capacity cannot compel surrender from an ideologically committed adversary, the prospects for cyber operations alone, which impose far less destructive and more reversible costs, are considerably weaker. The internet and related ICT have been instrumental in shaping public opinion, and hence national morale and ideology. From this view, if cyber coercion can work at all, it depends less on disrupting infrastructure and more on manipulating the informa-

tional environment that sustains an adversary's will to resist.

Russia's best cyber strategy for compelling Ukrainian surrender would prioritize informational operations targeting public morale over punitive infrastructure disruption, since punitive cyber attacks fail structurally due to their reversibility and inability to impose sustained costs (Borghard and Lonergan 2017). Such a strategy would be most effective if it was timed to exploit existing war exhaustion rather than attempting to create demoralization from scratch. Even then, this optimally designed strategy is very unlikely to succeed. Russia has conducted information operations against Ukraine since 2014, yet Ukrainian national identity has consolidated rather than fractured, consistent with the broader finding that coercive campaigns targeting morale tend to produce rally-around-the-flag effects rather than capitulation (Pape 2022).

2 Designing the Campaign

2.1 Dual Use Set up

As an analyst tasked with designing a cyber campaign to compel Ukrainian surrender, the central strategy would be to overcome the structural limitations of cyber operations as a tool of coercion. Punitive infrastructure attacks on power grids and financial systems face a fundamental problem of reversibility, as systems can be restored, patched, and hardened. This makes it difficult to exert pressure long enough for compellence to work(Borghard and Lonergan 2017). In addition, attacks on civilian infrastructure risk adverse effects for the attacker. Pape (2022) argues that coercive campaigns targeting civilian populations tend to generate rally-around-the-flag effects, strengthening national solidarity rather than eroding it. A viable cyber attack would seek to use cyber operations not to punish civilian populations but as a tool of information control. This paper proposes that the optimal cyber campaign would target telecommunications infrastructure in a dual-use manner: degrading

Ukraine's military coordination on one hand while shaping the civilian information environment. This should be complemented by direct cyber attacks on military infrastructure, targeting command and control systems, battlefield management networks, and logistics coordination to compound said operational disruption caused by telecommunications denial.

A cyber attack on telecommunications as the primary target exploits a key insight from coercion theory, where Biddle (2020) distinguishes between raising the costs of resistance and lowering the perceived benefits of continued fighting. For instance, an attack on financial systems or a power grid imposes only material suffering without giving Russia without any control over what Ukrainians may think. This in essence raises the cost of resistance while disregarding the other factor of lowering the perceived benefits of continued resistance. A telecommunications attack is unique, serving a dual purposes as it can degrade military communications and give access to Russia where it can exploit the information environment. Few other target set offers this dual function in achieving coercive objectives.

The timing of such an attack works best now. The onset of the war demonstrated unprecedented solidarity and willingness to fight. Contrast this with the current image of war strain and fatigue. Ukrainian society shows significant strain: public support for fighting until victory has collapsed from 63% to 24%, estimated 150,000 service members are AWOL, roughly two million citizens are evading the draft, and the ratio of volunteers to conscripts has fallen from 1:3 to 1:9 (meaning 90 percent of soldiers are conscripts)(Gallup 2025; Carnegie 2026; PBS 2026). Current morale conditions are therefore conducive to exploitation. The following subsections outline in detail how such an attack would be conducted.

2.2 Military Focus

The first component focuses on a cyber attack on the Ukrainian military's communications. Modern warfare encompassing a variety of domains requires close coordination between units,

drone operators, artillery, reconciles, command and control, etc. For this, modern militaries and specifically the Ukrainian military rely on military networks, commercial cellular infrastructure, Starlink satellite terminals, and messaging applications for battlefield coordination. Russian cyber operations would target all reachable communication systems in use by the Ukrainian military.

The goal is disruption, not destruction, which cyber attacks cannot easily achieve. This is to create an operational window tied specifically for the resumption of a summer offensive on Slavyaansk and Kramatorsk following an operational pause. During this pause, Ukrainian units cannot effectively act, acting as the denial mechanism as outlined by Pape (2022). This mechanism outlines how one may undermine the adversary's confidence in its ability to execute its own military strategy. Lindsay (2013) argues through the Stuxnet case that sophisticated cyber attacks require extensive intelligence preparation and resources that may outweigh the damage achieved. However, such insight in specialized industrial control software is difficult to extrapolate to telecommunications and network infrastructure. Telecommunications and Network infrastructure run on standardized architectures with known vulnerability classes and widely available exploit frameworks. The Human Intelligence (HUMINT) barrier is similarly lower, as Russia has large intelligence networks and sympathetic populations to leverage in a possible cyber attack which Iranian adversaries could not easily do at Natanz.

2.3 Psychological Impacts

2.3.1 Military Impacts

A cyber attack on the telecommunications and network infrastructure can have an important impact beyond purely traditional operational means. In the current conflict, both sides film flag raising videos, often with drones, and irrespective of ground control to demoralize opposing forces and sustain domestic morale. This was most critical in Ukraine's 2022

Kharkov offensive, where a rapid advance was amplified by footage of Ukrainian soldiers in captured towns. This contributed to the collapse of adjacent nearby towns, making an illusion of greater Ukrainian control than what was in actuality. Soldiers often retreated before receiving official orders. Battlefield information content is a weapon in itself, with units accepting tactical risk to produce propaganda footage.

A cyber campaign that recognizes and applies this insight would degrade Ukraine's ability to produce, disseminate, and consume this morale-sustaining content while flooding available channels with footage of Russian advances compounds operational disruption with psychological disruption. Soldiers who are demoralized, cannot communicate with command and cannot verify what happens in nearby sectors and are rather exposed to propagandized results are less capable of resisting an upcoming Russian offensive. Given that an estimated 150,000 service members are already AWOL and desertion is systemic (PBS 2026), even a marginal increase in psychological pressure on frontline units could accelerate an already critical retention crisis.

In addition to this,Ukrainian victories form a critical feedback loop between tactical success and external support. Ukrainian victories demonstrate proof of the value of further aid, first escalating from Javelins to tanks and ultimately F-16 fighter jets. Disrupting Ukraine's ability to publicize its victories through a cyber campaign on telecommunications infrastructure would weaken this cycle at its most visible link, potentially reducing international support that sustains Ukraine's capacity to fight.

2.3.2 Societal Impacts

However, the psychological impact of this campaign extends beyond the military but rather to the entire Ukrainian society. Biddle outlines a state's resolve as a single variable in a cost-benefit calculation (Biddle 2020), however, said states resolve is sustained by social trust,

institutional legitimacy, and shared national narrative. Each of these is under significant strain in Ukraine after four years of war and, although recently weakened, a long history of Russian influence. These factors make Ukraine particularly susceptible to an information cyber attack.

The most prominent outlook for this is the current mobilization crisis within Ukraine. Two million eligible citizens are actively evading military service, viral videos of "busification" or the forced conscription of citizens into minibuses have gone viral, eroding trust in government institutions (Carnegie 2026). This provides a susceptible ground for a cyber attack which instead of damaging infrastructure hijacks television, radio, internet, and other Ukrainian information means (rather than passively spreading propaganda) to raise the perceived costs of resistance. This will be done by pushing narratives of government corruption incompetence that target the institutional legitimacy that underpins the state's ability to demand sacrifice from its population. With the current conflict mirroring the dynamic of World War 1, the goal would be to achieve a similar collapse of morale in the rear as Germany had experienced.

The second tier would be in negotiations, where a cyber attack that spreads messages could promote the point of view of negotiations and even concessions within Ukraine. This is essentially an implementation of the chokepoint effect of Farrell and Newman (2019). Ukraine telecommunications and networks can serve as a chokepoint, where those who control the flow of information through these networks shape later political debate. A cyberattack as such should seek to create such a choke point or to deal a decisive blow to it. By flooding these networks with Russian propaganda in a more aggressive manner, Russia can gain leverage by lowering the perceived costs of continued resistance rather than raising its costs (Biddle 2020). However, in a distinct way the audience is not the Ukrainian military, but the wider public, who can exert political pressure when they see that the costs of continued

resistance outweigh the benefits of capitulation in some form or another.

Additionally, Hedgecock and Sukin (2023) find that covert operations generate less public support for retaliation than overt ones. For the campaign, this means that through manipulation rather than visible infrastructure destruction or disruption, by Hedgecock and Sukin's logic, would invoke less public backlash and reduce the likelihood of a nationalist rally-around-the-flag response. As a result this makes a potential cyber attack around communication means the most valuable and effective tool.

Finally, a cyber informational attack would also focus on weakening the informational foundation between Ukraine and its western partners. Nye (2017) describes how entanglement between states creates mutual stakes that sustain alliances. The current informational campaign would weaken this to the wider Ukrainian public by amplifying narratives of western abandonment, which given recent U.S. policies lends itself factual truth. This allows for an informational attack, where by hacking into communication networks it would be harder to dismiss transmitted content as purely Russian propaganda. The ultimate goal is to drive a wedge between Ukraine and her allies.

In general, the cyber campaign does not need to break the Ukrainian military or society outright or even be sustained. Borghard and Lonergan (2017) argue that a central weakness of cyber coercion is that its effects are temporary and reversible. Although a cyber attack may be easily reversed, by focusing on an information dimension, an idea that is spread cannot be so easily reversed. By using a cyber attack to promote information warfare in already fertile ground, a fragmentation which may raise the perceived costs of resistance rather than a single dramatic cyber attack on infrastructure is the most likely to fully coerce Ukraine.

3 Assessment: Likelihood of Success

The outlined campaign represents the strongest possible cyber coercive strategy given the current constraints of the domain and conflict. Targets large vulnerabilities while taking into account the weaknesses and failures of previous cyber attacks. Although with real war exhaustion in Ukraine and meeting the theoretical requirements of Biddle's for compliance, this cyber attack is very unlikely to compel Ukraine into full surrender. Large obstacles still remain: the structural limitation of cyber coercion, the strength of Ukraine information environment, and the gap between war fatigue and the acceptance of Russian maximalist demands.

3.1 Difficulties of operational effects

The military denial prong proposes creating an operational window for a Russian offensive on Slavyansk and Kramatorsk by disrupting Ukrainian military communications. However, this approach faces several problems. First, the Ukrainian military has been operating in a contested cyber environment and has likely developed countermeasures for a widespread attack. Ukraine likely has fallbacks and recovery systems that can negate or make a wide spread attack difficult. Although the attack plan argued that an attack would be unlike Stuxnet (against very specific ICS), this cuts both ways, where standard architectures have standard recovery procedures and fixes are more widely available. This fits in with Slayton (2017) arguing that the cyber offense-defense balance favors the defender over time precisely because each attack teaches the defender how to respond faster, with Ukraine having now over a decade of experience in defense. Furthermore, given the weaponized interdependence of the internet and telecommunications, Russia would have to attack infrastructure belonging to third party entities, such as Starlink. This means that other states through which the attack may route can take advantage of chokepoint or panopticon effects to stop or inform Ukraine in advance of an incoming cyber attack.

Furthermore, the coordination of cyber and ground offensives presents significant challenges. Borghard and Lonergan (2017) identify a fundamental problem with synchronizing cyber- and kinetic operations. A sophisticated cyber attack of this magnitude requires deep planning and preparation where the cyberattack can be discovered or disrupted before it begins or when a ground offensive may begin. This is made more likely by Ukraine's cyber institutional strength, where Kostyuk (2021) reinforces this by demonstrating that states with mature public cyber institutions, which Ukraine has built with Western support since 2014, are better at detecting intrusions during the preparation phase.

3.2 The Informational Track Fails Against Ukraine

Although promising, the information aspect of the cyber attack still faces difficulties in the context of Ukraine. Although several smaller information choke points exist (radio, television, etc.), they are still multiple and more fragmented than a network like Swift as outlined by Farrell and Newman (2019). Hijacking a variety or even a plurality of these networks does not guarantee a chokepoint of the strength outlined as with swift, thus bringing into question if such a choke point analogy is possible on an information space. Another possible issue is backlash, where a covert operation as argued would generate less backlash. Although less likely than other cyber and definitely kinetic means, it is still possible that a backlash would be created from such an attack that may blunt, reverse, or even damage Russia in the information space domain. Passively amplifying existing grievances through bot networks and social media manipulation could remain covert, but the aggressive hijacking of Ukrainian broadcast channels that the campaign describes could trigger exactly backlash Pape (2022) predicts. Furthermore, while morale has suffered, the resolve to the Ukrainian war is likely still strong, having grappled with Russian information warfare for more than a decade now with limited gains for Russia and a Ukrainian policy of de-russifying its information space. Given the possibility of backlash, Ukraine's resilience, and its own information

space strength, this makes the information space attack not enough to force a Ukrainian capitulation.

3.3 The Maximalist Demand Problem

Finally, even with a fully successful information and operational cyber attack, the question remains whether this alone can raise the costs of resistance and lower the benefits of resistance when the ultimate goal of compellence is a total capitulation. This raises not a tactical concern, but a structural one: whether the category of the instrument can match the scale of the demand before it.

With Biddle's framing of coercion theory as a cost benefit analysis between the costs of resistance exceed the costs of concession. However, when quantifying this, the price of capitulation is viewed as existential to the Ukrainian state, sovereignty, and even national identity; it is difficult to imagine if such a high price can be lowered in any meaningful way as the cyber attack aims to. The outcomes of a cyber attack are not binary. Combined with war exhaustion, this could even increase perceived costs enough to extract meaningful concessions such as Russian annexation of the Donbass, recognition of Novorossiya, Ukrainian demilitarization, neutrality, or even changes in language policy. Although sharply harsh concessions, each one falls short of total capitulation as when the price of concession is the end of the Ukrainian state, its sovereignty, and national identity itself, it is difficult to imagine any cyber operation lowering that price in a meaningful way. A state can cede territory and endure. It cannot accept its own extinction and endure. This is the threshold that cyber coercion, regardless of design, cannot cross alone.

4 Conclusion

Even an optimally designed cyber campaign, combining both military telecommunication, command and control, and informational defeatism, is unlikely to compel Ukrainian surrender alone. This is due in part to the diminishing returns from reversibility, Ukrainian defensive expertise, an unfavorable information space, and the risk of backlash. However, most importantly and structurally, the demand for total capitulation is a threshold that no cyber instrument alone can cause. While meaningful concessions can be extracted, the extinction of a state is something incredibly difficult to force, being the ultimate price.

However, the current Russian doctrine recognizes this implicitly or explicitly. New generation warfare, or "the Gerasimov Doctrine", treats cyberspace as a supporting tool rather than a standalone coercive instrument. The findings of this essay are that Cyber alone is not a sufficient tool of coercion as its effects are reversible, and the information track has difficulties in overcoming an ideologically resilient adversary.

The situation that makes such a strategy viable in the first place, four years of full scale war, demonstrates that cyber cannot alone force compellence, but rather kinetic actions that overwhelm the enemy. However, cyber can still amplify, accelerate, and exploit the conditions that conventional warfare creates, but it cannot generate the foundational costs that make coercion possible. Cyber has a coercive ceiling as outlined by Lawson and must be viewed differently, as outlined by both Russia's new generation warfare and Rovner (2019) who argues that cyber functions most effectively as an intelligence contest rather than a decisive strategic instrument. This essay confirms that, while a powerful tool, cyber is not strong enough to force capitulation despite offering serious promise.